SYSTEM PROGRAMMING ROADMAP

I. STATUS ABBREVIATIONS

- **C** - Complete
- **I** - Incomplete
- **O** - Ongoing
- **S** - Soon

II. SUMMARY

Sno.	Study Area	Status	Requirement
1	C Programming	**C**	Making software
2	C++ Programming	**C**	Standby skill
3	Data Structures & Algorithms	**C**	Optimization
4	Operating System	**C**	Kernel architecture
5	Bash Scripting	**C**	Automation
6	Computer Network	**C**	TCP/IP stack
7	Software Engineering	**C**	Project management
8	Compiler Design	**O**	Command-line projects
9	x86 Assembly	**I**	Optimization & hardware
10	Tools/Platforms	**C**	Smoother development
11	System-Level Calls	**O**	Kernel system calls
12	Device Drivers	**S**	Hardware device interaction
13	Bare Metal Programming	**S**	Vendor specific projects

III. STUDY AREAS

1. C Programming (C):

Sno.	Topic/Chapter	Status
1.1	Memory management	C
1.2	Pointers	C
1.3	String operations	C
1.4	Structures & Unions	C

Sno.	Topic/Chapter	Status
1.5	Data structures	C
1.6	File handling	C
1.7	Preprocessor directives	C
1.8	Linking & Libraries	C

2. C++ Programming (I):

Sno.	Topic/Chapter	Status
2.1	Classes & objects	C
2.2	Memory management	C
2.3	File handling	C
2.4	Vectors	C
2.5	Unordered maps	C
2.6	Generic	C
2.7	Multithreading	C
2.8	Smart pointers	C
2.9	Lambda expressions	C

3. Data Structures & Algorithms (C):

Sno.	Topic/Chapter	Status
3.1	Asymptotic notations	C
3.2	Abstract data types	C
3.3	Sorting algorithms	C
3.4	Linked lists	C
3.5	Trees	C
3.6	Graphs	C
3.7	Hashing	C
3.8	Turing machine	C

4. Operating System (C):

Sno.	Topic/Chapter	Status
4.1	Kernel	C
4.2	Process management	C
4.3	Threading	C
4.4	CPU scheduling	C
4.5	Deadlocks	C
4.6	Memory management	C
4.7	Virtualization	C
4.8	I/O systems	C
4.9	Disk management	C

5. Bash Scripting (C):

Sno.	Topic/Chapter	Status
5.1	Arguments & arrays	C
5.2	Functions	C
5.3	Traps	C
5.4	Scripting elements	C
5.5	Base conversions	C

6. Computer Network (C):

Sno.	Topic/Chapter	Status
6.1	Network topologies	C
6.2	Networking devices	C
6.3	Data link layer	C
6.4	Network layer	C
6.5	Transport layer	C
6.6	Application layer	C

7. Software Engineering (C):

Sno.	Topic/Chapter	Status
7.1	Software development lifecycle	C

Sno.	Topic/Chapter	Status
7.2	COCOMO model	C
7.3	Software metrics	C
7.4	Project planning	C
7.5	Testing & validation	C

8. Compiler Design (O):

Sno.	Topic/Chapter	Status
8.1	Deterministic finite automata	C
8.2	Regular expressions	C
8.3	Context free grammar	C
8.4	Parsing	C
8.5	Syntax Directed Translation	C
8.6	Code Case Studies	
8.7	Symbol tables	
8.8	Storage	
8.9	Error detection	
8.10	Code generation	
8.11	Code optimization	

9. x86 Assembly (O):

Sno.	Topic/Chapter	Status
9.1	Family of x86 chips	C
9.2	Assembly syntax comparisons	
9.3	Inline assembly	
9.4	Bootloading	
9.5	x86 CPU architecture	

10. Tools/Platforms (I):

Sno.	Topic/Chapter	Status
10.1	GCC	C

Sno.	Topic/Chapter	Status
10.2	G++	C
10.3	Git	C
10.4	GDB/Valgrind	C
10.5	Linux	C
10.6	CMake	C

11. System-Level Calls (O):

Sno.	Topic/Chapter	Header	Status
1.1	Standard library	`<stdlib.h>`	
1.2	UNIX standards	`<unistd.h>`	
1.3	File control	`<fcntl.h>`	
1.4	Singal handling	`<signal.h>`	
1.5	Multithreading	`<pthread.h>`	C
1.6	Socket Programming	*Multiple*	C
1.7	Memory mapping	`<sys/mman.h>`	
1.8	Direct system call	`<sys/syscall.h>`	

12. Device Drivers (S):

Sno.	Topic/Chapter	Status
12.1	Running modules	
12.2	Character drivers	
12.3	Concurrency & race conditions	
12.4	Time & delays	
12.5	Memory allocation	
12.6	Hardware bridging	
12.7	Interrupt handling	
12.8	PCI drivers	
12.9	USB drivers	
12.10	Block drivers	
12.11	Network drivers	
12.12	TTY drivers	

13. Bare Metal Programming (S):

Sno.	Topic/Chapter	Status
13.1	Development environment	
13.2	x86 Bare metal	
13.3	Booting & setup	
13.4	Testing & debugging	
13.5	Shell development	

IV. Statistics

1. Subject statistics:

$$Subjects\ comepleted\ =\ \frac{8}{13}\ =\ 61\%$$

$$Subjects\ incomeplete\ =\ \frac{1}{13}\ =\ 8\%$$

$$Subjects\ ongoing\ =\ \frac{2}{13}\ =\ 15\%$$

$$Subjects\ coming\ =\ \frac{2}{13}\ =\ 15\%$$

2. Topic statistics:

$$Topics\ comepleted\ =\ \frac{64}{98}\ =\ 65\%$$

$$Topics\ incomeplete\ =\ \frac{34}{98}\ =\ 34\%$$